Big Data Analysis of Online Retail Transactions Using Apache Spark

IT 462 – Big Data Systems
Group 5
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Problem and Motivation

Context

Retail businesses collect large transactional data but often lack clear insight into customer behavior, loyalty, and inactivity.

Problem

Identify meaningful customer segments and potential churn-risk customers using RFM features from unlabeled retail transaction data.

Target User

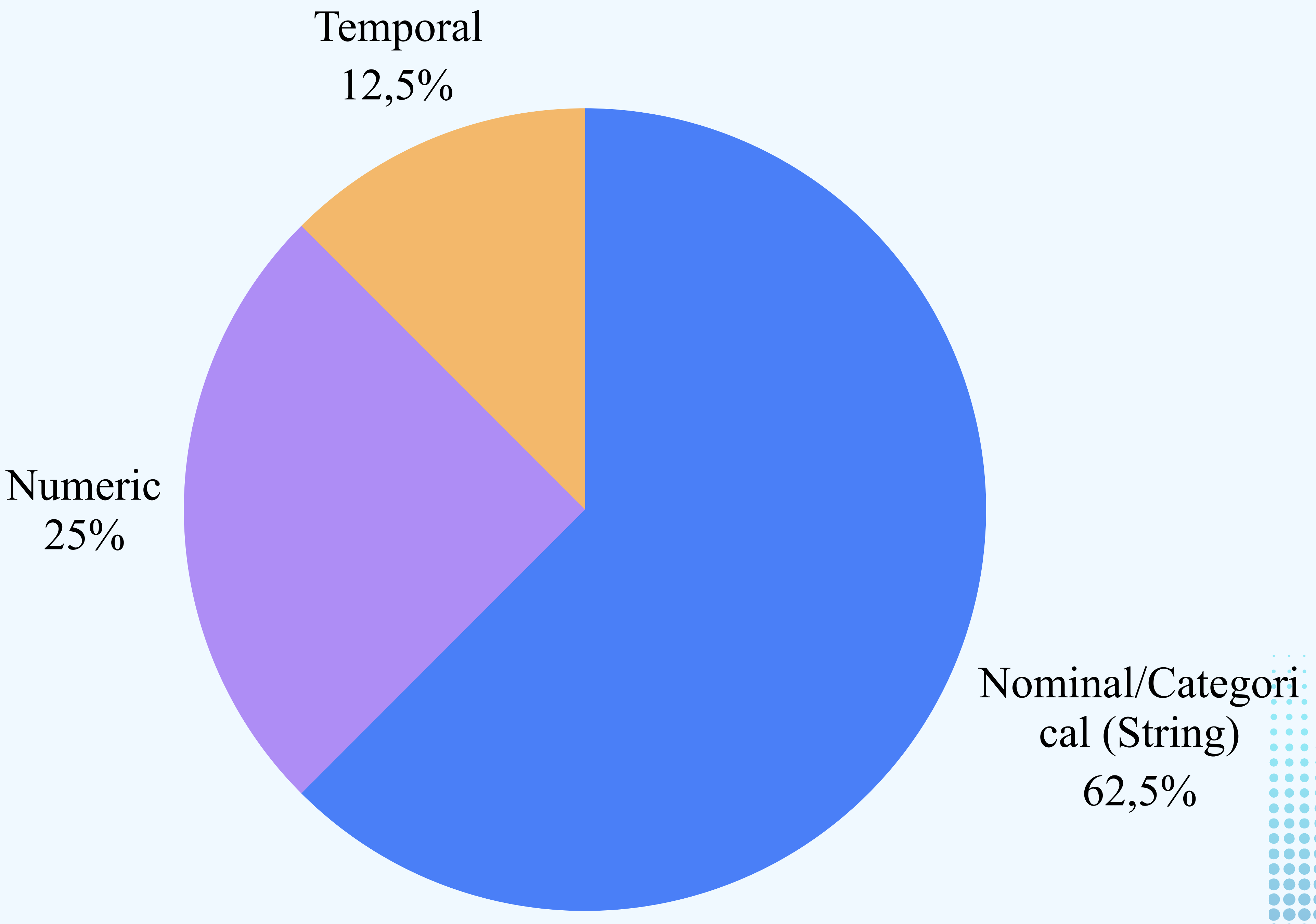
Marketing teams, business managers, and customer retention analysts.

Why it matters

The results help businesses detect valuable and at-risk customers, compare purchasing patterns, and design targeted retention strategies.

Dataset Overview

- Online Retail II, from Kaggle
- 1,067,371 rows and 8 columns.
- E-commerce and Online Retail Analytics



Data Quality & Preprocessing

Main Data Quality Issues

- Missing values were found mainly in CustomerID and Description.
- Invalid transaction records were detected, such as Quantity ≤ 0 and UnitPrice ≤ 0.
- Duplicate records and outliers in Quantity .

Issue	Before	After
Missing CustomerID rows	243,007	0
Missing Description	4382	Replace with “Unknown”
Invalid Quantity/Price	18,815	0
Duplicate records	26,124	0
Outliers	51,119	0
Final dataset size	1,067,371	728,306

Key Preprocessing Steps

- Removing rows without CustomerID and replacing missing Description with “Unknown.”
- Removed invalid and duplicate transactions to improve data quality.
- Selected relevant features and transformed the data into customer-level RFM features for analysis.

RDD Exploration: Churn Candidate Identification

Operations:

map → Extract customer RFM features from each row

filter → Identify customers with high churn risk

take → Retrieve the top inactive customers

Operations:

```
val customerFeatures = rows.map(row =>
(row(0), row(1), row(3).toInt, row(4).toInt, row(5).toDouble)
)

val churnRDD = customerFeatures.filter {
case (_, _, recency, frequency, _) =>
recency > 90 && frequency < 5
}
```

Sample Result:

CustomerID	RecencyDays	Frequency
13526	738	2
17056	738	1
12636	738	1

RDD Exploration: Country-Level Aggregation

Operations:

map → Convert customer records into country-based pairs

reduceByKey → Aggregate churn metrics per country

mapValues → Compute average spending and churn ratio

Operations:

```
val countryRDD = customerRDD.map(x =>
(x._1, (1, x._3, x._4)))

val result = countryRDD.reduceByKey((a,b) =>
(a._1+b._1, a._2+b._2, a._3+b._3))
```

Sample Result:

Country	High-Risk	Churn Ratio
Australia	5	333
Brazil	1	500
Canada	2	500

SQL Operations

- The preprocessed dataset was loaded into a Spark DataFrame and registered as a temporary SQL view named:

```
df.createOrReplaceTempView("customers")
```

- Spark SQL queries were used to analyze customer purchasing behavior and identify churn-risk segments.

SQL Operations

Key Analytical Queries

- Country-based customer value analysis
- Detection of inactive high-value customers
- Churn percentage analysis by country
- Customer risk segmentation using CASE statements
- Ranking high-risk customers using window functions and CTEs

Key Insights

- High-value inactive customers were identified as potential churn-risk customers.
- Customer purchasing behavior varied significantly across countries.
- Customers with high recency and low frequency showed higher churn risk.
- RFM features provided strong behavioral patterns for clustering.

Machine Learning Operations

Problem Type

Unsupervised learning
(Clustering)

Features Used

- RecencyDays
- Frequency
- Monetary (RFM Model)

Models Used

1. K-Means
2. Gaussian Mixture Model (GMM)

Training Setup

- Train/Test Split: 70% / 30%
- Random Seed: 42
- Evaluation Metric: Silhouette Score

Machine Learning Operations

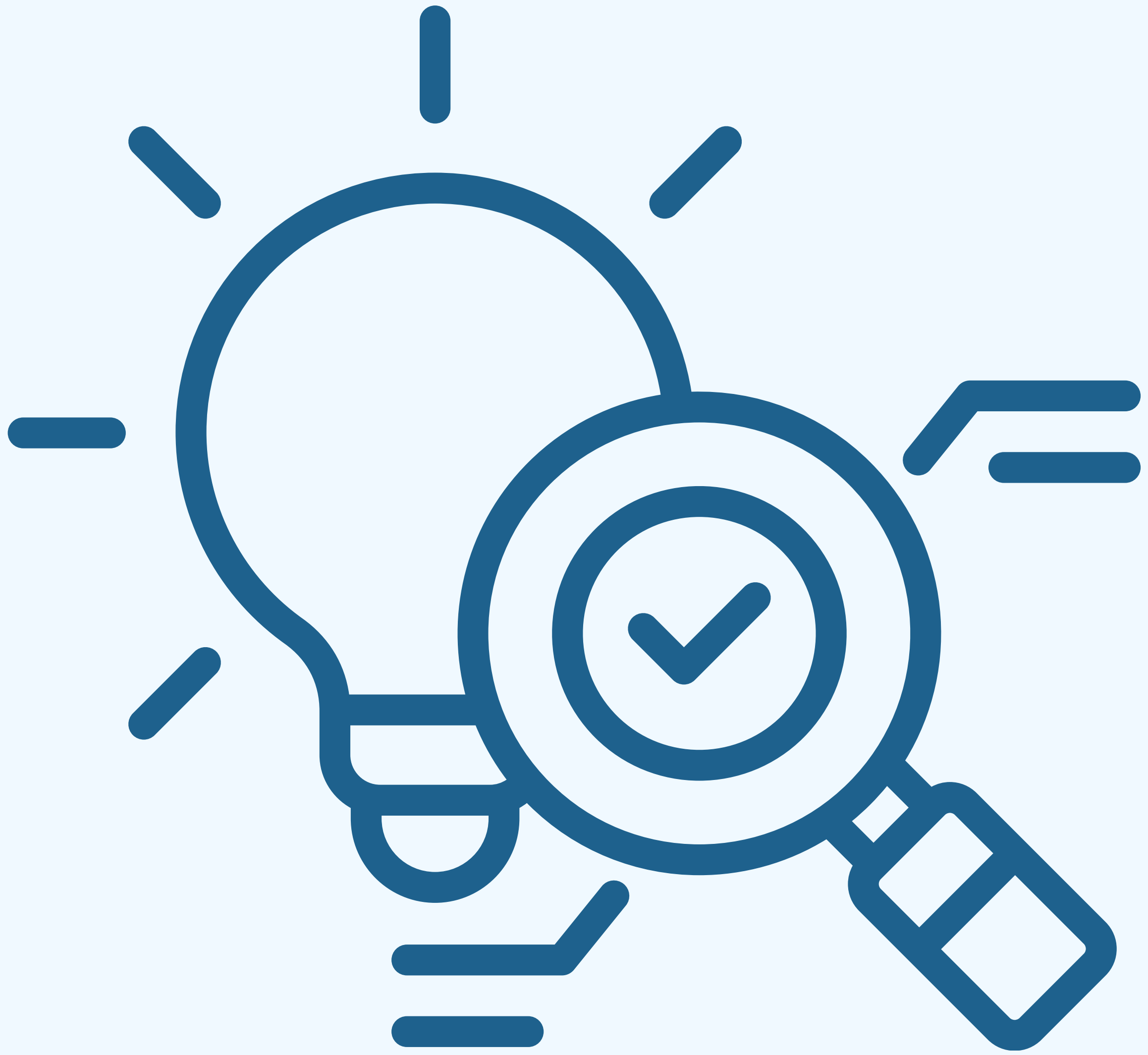
Results

Model	Best K	Silhouette Score
K-Means	2	7.232
GMM	2	6.419

Machine Learning Operations

Final Findings

- K-Means achieved the best clustering performance.
- The model successfully separated:
    1. loyal high-value customers
    2. inactive low-value customers
- Cluster analysis helped identify potential churn-risk customers.



5 Marketing & Promotion Strategy

Digital Marketing

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Content Strategy

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Campaign Optimization

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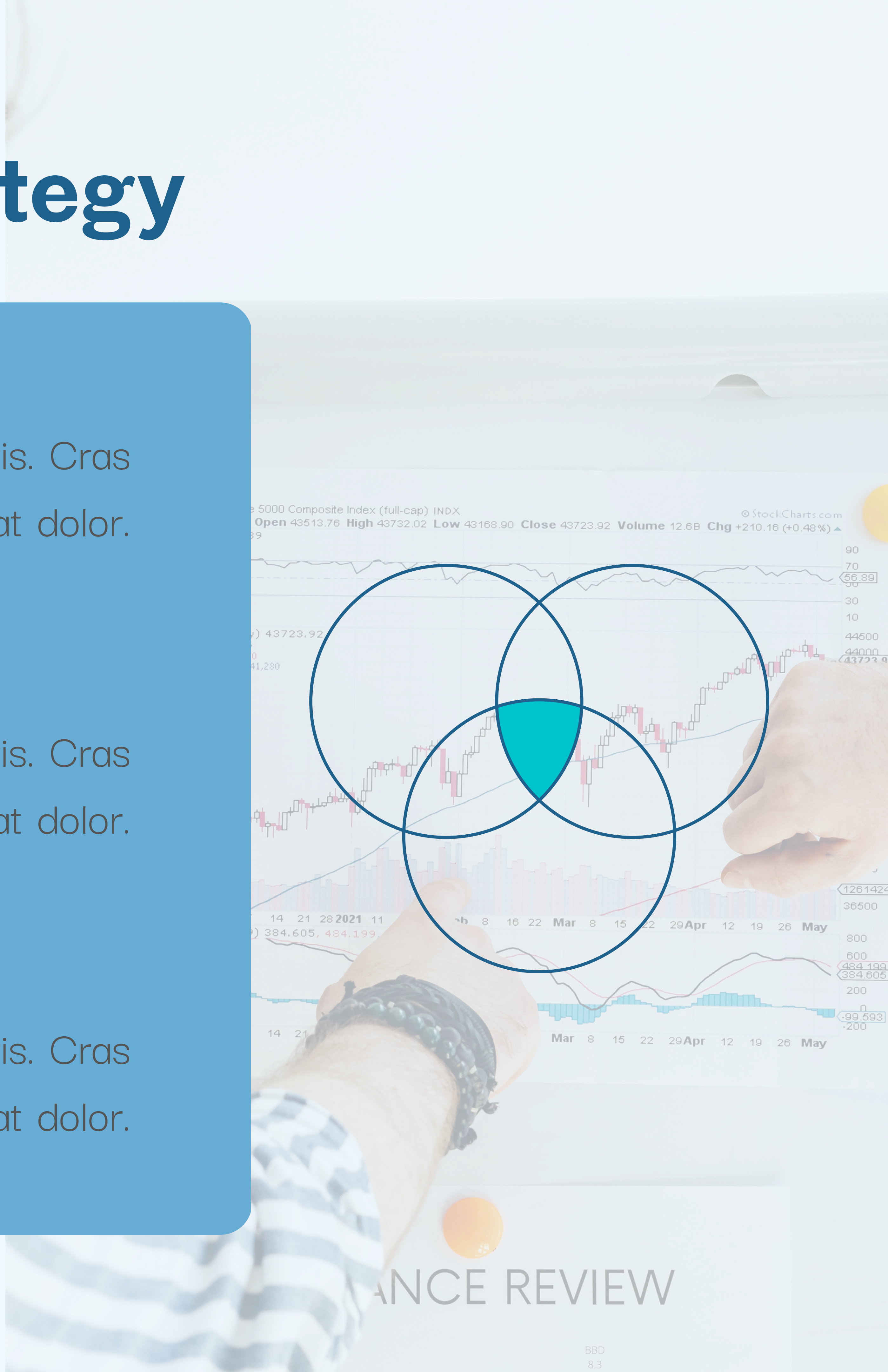

6 Inventory & Supply Chain Planning

Inventory Control

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Supply Chain

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Demand Forecasting

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7 Financial Performance Metrics

Sales & Revenue

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Profit & Costs

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Cash Flow

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Risk Assessment & Mitigation Plan

Risk Identification

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Risk Mitigation

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9 Strategic Roadmap & Next Steps

Strategic Milestones

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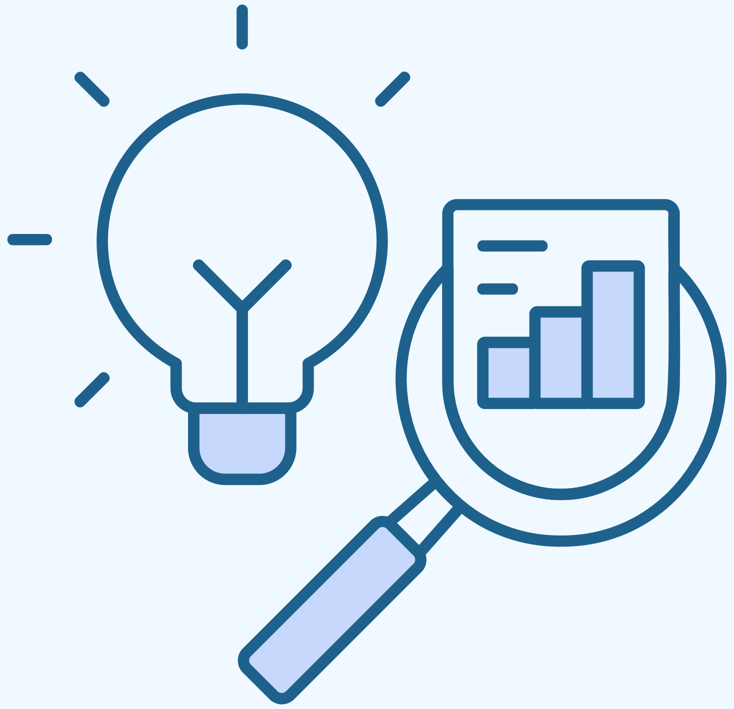

Risk Mitigation

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Much appreciated.